MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
SECOND SEMESTER EXAMINATION
(2024 ADMISION ONWARDS)
MG2DSCZIM102 – Molecules of Life

Duration: 1.5 hrs Maximum Marks: 50

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A
Fill in the blanks
Answer **All** Questions
Each question carries **1** mark

1.Molecules have only 1 potential sugar group are called…….. [Remember] [CO1]
2.All proteins are……of aminoacids [Understand] [C02]
3.Double helical model of DNA was proposed by…… [Remember] [CO5]
4.Lipids are esters of fatty acids with …… [Remember] [CO4]
5.The pH at which amino acid carry no net charge is called…… [Understand] [C02]
6 ….……..is the reserved carbohydrates in plant kingdom [Remember] [CO1]
7.The sugar in DNA is…… while RNA is ribose. [Remember] [C05]
8.The basic unit of nucleic acid is…….. [Remember] [CO5]
9.In carbohydrates change in specific rotation with time is called…….....[Remember] [CO1]
10.Hemoglobin is a……protein found in red blood cells [Remember] [CO3]
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[1 x 10 = 10]

Part B
Short Questions
Answer10 **Q**uestions
Each question carries 2 marks

11. Differentiate between monosaccharides and disaccharides [Understand] [CO1]
12. Outline the structure of maltose [Understand] [CO1]
13. Write short note on tertiary structure of proteins. [Apply] [CO3]
14. Define saponification number [Understand] [CO4]
15. Discuss the importance of mucopolysaccharides? [Understand] [CO1]
16. Write short note on denaturation and renaturation of proteins. [Understand] [CO2]
17. State chargaff's rule? . [Understand] [CO5]
18. Differentiate between essential and non essential amino acids. [Understand] [CO2]
19. Define mutarotation [Understand] [CO1]
20. Disuss the physiological importance of phospholipids. [Understand] [CO5]
21. Discuss the characteristic features of nucleic acid? [Understand] [CO5]
22. Illustrate the structure amino acid glycine? [Apply] [CO2]

[2 x 10 = 20]

Part **C**

Short Essay Type Questions

Answer 5 Questions

Each question carries 4 marks

23. Write an essay on classification of carbohydrates. [Understand] [CO1]
24. Explain Watson and Crick model of DNA. [Understand] [CO5]
25. Compare and contrast primary and secondary structure of protein. [Apply] [CO3]
26. Define isoelectric pH.Explain classification of aminoacid based on nutritional requirement. [Understand] [CO2]
27. Explain chemical constants of fatty acid? [Apply] [CO4]
28. Discuss the significance of Lecithin and cephalin. [Understand] [CO5]
29. Write an essay on homopolysaccahrides and heteropolysaccharides [Understand] [CO1]

[4 x 5 = 20]